Does aging microplastics alter how particles affect the physiology of a freshwater copepod

The Zooplankton Ecology & Adaptation Lab At UMass Lowell

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Introduction

- Microplastics are increasing in abundance in aquatic systems
- Changes in food quality and quantity
- Copepods are important, highly selective aquatic grazers
- Biofilms can form on surface of microplastics (Nutritional benefit? Increased exposure via reduced selectivity?)

Central Question

Does microplastic aging alter how these particles affect the physiology of a common freshwater copepod?

Hypotheses

1. Exposure to microplastics will **decrease** copepod thermal limits.
2. Thermal limits of fed copepods will be **higher when exposed to aged than un-aged** microplastics.
3. Because of the nutritional content of biofilms, this **pattern flips for starved copepods**.

Preliminary Results

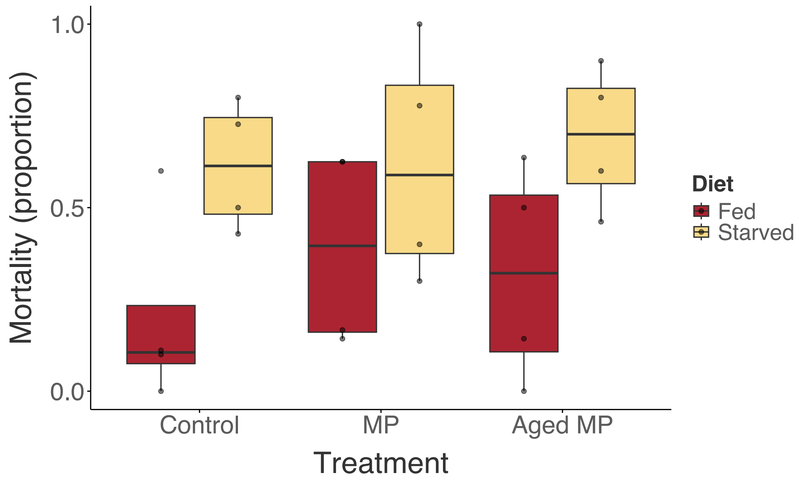


Figure 1: Proportional mortality in the different treatments. Survival was measured after one week. Each point shows mortality in one of the replicates, but replicates across experiments are pooled together. Mortality was consistently higher in the starved treatments.

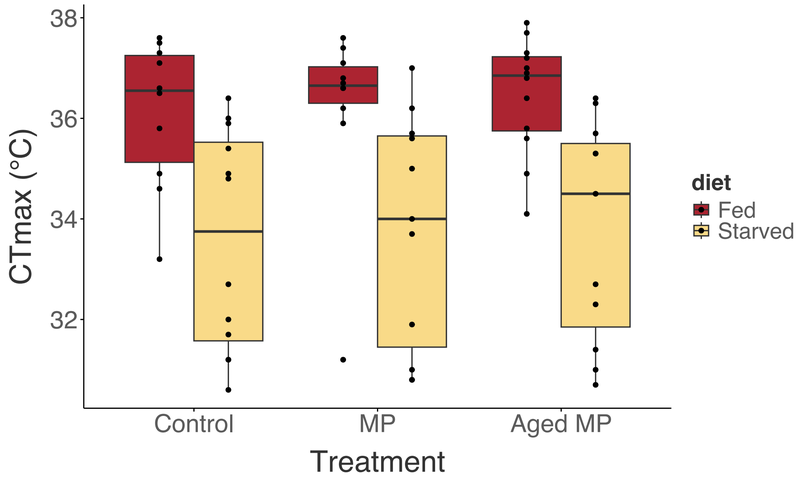
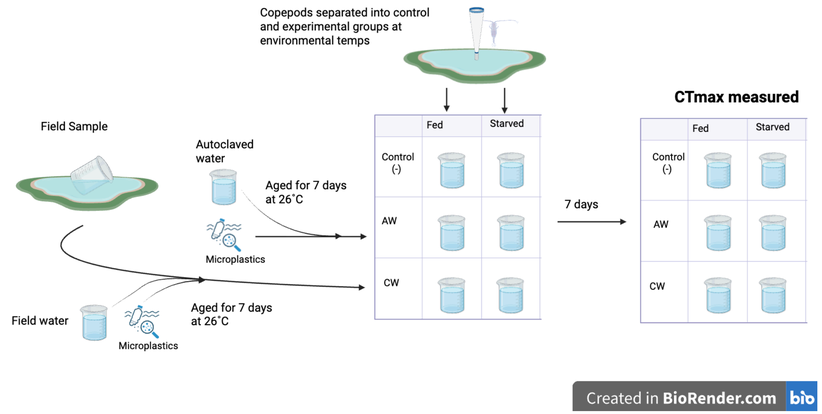


Figure 2: Upper thermal limits for copepods in the different treatments. CTmax was measured after one week. Each individual point represents the observed upper thermal limit of an individual copepod. Data from the two experimental replicates is pooled.

Methods



Conclusion

- Starvation has major effects on both survival and thermal limits
- Realistic microplastic concentrations have minimal effects on thermal limits, but may influence survival
- Survival seems to be highly variable across weeks

Next Steps

- Additional experimental replicates
- Dose-dependence of effects

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